

Solution-Operator Semi-Discretization: General-Order Multiplication-Free Stability Analysis of Time-Periodic Delayed Systems and the Accuracy–Sparsity Trade-Off

Daniel Bachrathy^{1,*}

¹Department of Applied Mechanics, Faculty of Mechanical Engineering,
Budapest University of Technology and Economics,
Műegyetem rkp. 3., H-1111 Budapest, Hungary

*Corresponding author: bachrathy@mm.bme.hu, Tel.: +36-1-463-1369
ORCID: 0000-0002-7268-2999 [verify before submission]

Article type: Original research article

Funding: The research leading to these results was supported by the Hungarian Scientific Research Fund (NKFIH-152125).

Declaration of conflicting interests: The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability: All methods, benchmark scripts and figure-generation pipelines are openly available in the Julia package SOSD.jl at <https://github.com/bachrathyd/SOSD.jl>; every figure of the manuscript can be regenerated from the repository.